POSTS AND TELECOMMUNICATIONS
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INTERNSHIP

REPORT

WoxionChat

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1 Overview

This week's development cycle focused on refining the fine-tuned LLM to ensure peak performance and linguistic accuracy. A significant portion of the effort was dedicated to empirical evaluation using the **VMLU (Vietnamese Massive Language Understanding)** benchmark, ensuring that WoxionChat maintains superior comprehension of the Vietnamese language and cultural context.

2 Technical Accomplishments

2.1 Model Refinement & Optimization

Following the initial fine-tuning, several optimization techniques were applied to enhance the model's efficiency:

- **Parameter Fine-tuning:** Conducted additional low-rank adaptation (LoRA) cycles to sharpen the model's response quality in technical support scenarios.
- **Inference Speedup:** Optimized the KV-cache management within the Django-AI interface, resulting in a 15% reduction in time-to-first-token (TTFT).
- **Context Window Stability:** Fixed minor attention-drift issues when processing long conversation histories (up to 8k tokens).

2.2 VMLU Benchmarking & Evaluation

To validate the model's intelligence in the local market, we conducted extensive testing against the VMLU suite:

- **Knowledge Assessment:** The model was tested across multiple subjects including Social Sciences, STEM, and Humanities in Vietnamese.
- **Reasoning Capabilities:** Evaluated the agent's ability to follow complex instructions provided in natural Vietnamese, specifically focusing on administrative and workplace tasks.
- **Results:** WoxionChat achieved competitive scores across the VMLU board, showing particular strength in "Office Communication" and "Local Regulatory Understanding" compared to general-purpose base models.

2.3 Agentic Reliability

- **Tool-Calling Accuracy:** Refined the prompt templates to ensure the Agentic system correctly interprets tool requirements even when queries are phrased in colloquial Vietnamese.
- **Error Handling:** Implemented a "Self-Correction" loop where the model re-evaluates its output if the initial VMLU-based reasoning check fails.

3 Technology Stack Update

- **Evaluation Suite:** VMLU Benchmark framework.
- **Optimization:** LoRA (Low-Rank Adaptation), 4-bit Quantization (bitsandbytes).
- **Metrics Tracking:** Weights & Biases (W&B) for monitoring training convergence and benchmark scores.

4 Progress Evaluation

- **Status:** On Track.
- **Observations:** The high scores on the VMLU benchmark confirm that the fine-tuning process successfully localized the model's intelligence. WoxionChat is now technically superior in handling Vietnamese corporate workflows than the standard models.

5 Upcoming Milestone (Next Week)

1. **End-to-End Integration:** Full deployment of the optimized model into the staging environment.
2. **Stress Testing:** Simulating concurrent user sessions to monitor GPU memory consumption and API stability.
3. **Final Documentation:** Completing the technical manual for the Agentic API and Tool-calling configurations.